

Marine biologists have been sounding the alarm about ocean pollution since the 1950s. Up until then, many scientists believed that the ocean was large enough to dilute any human waste and protect its animal inhabitants. Over the past few decades, researchers have shown that man-made pollution directly harms marine life---but there's still far more work to do. Biologists have only studied the effects of harmful chemicals on a handful of ocean-dwelling animals, leaving the influences of contaminants on the rest of sea life largely unknown. Now, scientists are trying to close that knowledge gap in whales and dolphins. A study out last week in *Frontiers in Marine Science* surveyed 83 animals stranded on shores across the southeastern US from 2012 to 2018 to investigate how toxic chemicals pervade these animals' bodies, and in what quantities. The 83 animals represent 11 species, some of which scientists know little about as human sightings are incredibly rare.

Every time a marine mammal is found dead and stranded in their area, Annie Page-Karjian, a clinical veterinarian at Florida Atlantic University who specializes in marine animals and lead author of the paper, heads out to do a necropsy---the animal equivalent of an autopsy. "We take hundreds of samples from their organs and we do a full pathological analysis," she says. During necropsies, Page-Karjian looks for damage to the animal's organs, identifies toxic chemicals in their tissues, and flags any diseases the animal may have been suffering from. Whales and dolphins are ideal subjects for marine toxicology research, she says, because of what they eat and how long they live. "Cetaceans are apex predators," Page-Karjian says. They're at the tops of their food webs, which means that the chemicals and pollutants various ocean species eat or absorb will amass upwards as predators eat prey---a phenomenon known as bioaccumulation. "These whales and dolphins accumulate compounds within their body organs and blubber over the course of their long lives, and they provide a snapshot of what's going on in the marine environment."

And that snapshot, Page-Karjian's team found, is alarming. One Gervais' beaked whale, a species that humans rarely see, was found on a beach in Sebastian, Florida in 2017---it had the highest liver concentration of arsenic of any marine mammal to date. This is especially shocking, Page-Karjian says, when you consider that their normal habitat is so far removed from human populations. Two bottlenose dolphins---one found in North Carolina in 2012 and the other in Florida in 2018---both had liver mercury concentrations that were some of the highest of any animal researchers have studied. Further, one white-beaked dolphin they studied had two different kinds of cancer in two organs: in its testicles and kidneys. Cancer in any marine mammal is rare, so finding two different kinds in one animal was an unpleasant surprise, says Page-Karjian. They have a suspected culprit, too. "This animal had the highest BPA concentration, [a known] carcinogen, of any of the animals that we looked at." The BPA level was also higher than any other studies done in the past have published as well, though that data is very limited. Page-Karjian says that these results are concerning, not just for whale and dolphin populations, but for humans as well. "Transfer of contaminants and bio-accumulation [are not] isolated to animals," she explains, "it also happens in humans."

1. According to the text, which of the following statements is true?
 - a. Scientists were concerned about ocean pollution
 - b. There are 11 rare animals being investigated
 - c. Bio-accumulation doesn't occur in humans
 - d. Cetaceans are alpha predators
 - e. Two dolphins found in Florida had the highest liver mercury concentrations

2. According to the last paragraph, it can be concluded that...
 - a. Ocean pollution harms sea creatures
 - b. Transfer contaminants and bio-accumulation happened in ocean-dwelling animals
 - c. Beaked whale's liver had huge amount of arsenic
 - d. Whale and dolphins are chosen to be investigated because they are apex predators
 - e. Cancer in marine mammal is rare

Climate change can have profound impacts across ecosystems, but rising average temperatures are just one factor among many driving those repercussions. A new study published in late September in *Global Change Biology* found that nighttime temperatures are increasing at a faster rate compared to daytime temps in most land areas across the Earth.

"Climate change is already messing things up," says Daniel Cox, an ecologist at the University of Exeter and lead author of the study. "But the 24-hour asymmetry is adding an extra dimension of complexity." But Cox says this is the first study of temperature asymmetries to cover all global lands. Experiments with grasshoppers and spiders, for example, have shown that the time of the day at which heating occurs can tip the ecological balance. In a 2017 study, researchers found increased daytime warming led spiders to seek cover earlier in the day, enabling grasshoppers to munch away at plants without fear, affecting plant growth. Conversely, the spiders hunted the grasshoppers more fiercely when nighttime temperatures warmed, possibly reducing the insect's numbers. These kinds of effects can ripple across a larger ecosystem, with potential impacts for plant communities, wildlife, and agriculture.

Nighttime ecology is particularly understudied, with most research focused on the daytime activities of organisms. That's why Cox was interested in understanding temperature asymmetries across the globe. Revealing these differences can be a stepping stone toward seeing how nocturnal activities are faring under climate change. To find out, Cox and his team mapped 35 years of data on temperature, cloud cover, humidity, and precipitation. For each of the pixels of land area on the global maps, they looked at how the maximum daytime and minimum nighttime temperatures changed over time.

On a global level, nights are heating up more than days. Almost twice as much area has seen a greater temperature increase at night compared to during the day. This change appears to be closely tied to shifts in cloud cover; where cloud cover increased, so did nighttime temperatures. "We think that cloud cover is probably the main driver," says Cox. In wetter regions, warming temperatures are causing more water to evaporate, leading to more clouds. Clouds cool during the day by blocking the sun, but at night they trap heat close to the ground. In general, wetter regions over time got cloudier and thus warmer at night. Meanwhile, a smaller portion of already-arid land grew drier as temperatures increased the most during the day.

3. Which of the following is false?
 - a. The rising average temperature is one of the climate change's effects
 - b. Daniel did the first ever study of temperature asymmetries to cover all global lands
 - c. The daytime ecology is literally overstudied, leaving the nighttime ecology
 - d. The team plotted more than 30 years of temperature, cloud cover, humidity, and precipitation data
 - e. The cloud cover has nothing to do with the change in nighttime's temperatures
4. To put in another word, we can say that this text talks about...
 - a. There are reduce in the grasshoppers' numbers
 - b. One of climate change's repercussions is rising average temperature
 - c. Wetter regions got cloudier at daytime and warmer at night
 - d. Nowadays climate change is really messed up
 - e. The shift in daily temperature influences how plants and animals cope with climate change

This passage is adapted from Will Douglas Heaven, "DeepMind says it will release the structure of every protein known to science." MIT Technology Review.

¹Back in December 2020, DeepMind took the world of biology by surprise when it solved a 50-year grand challenge with AlphaFold, an AI tool that predicts the structure of proteins. ²Last week the London-based company published full details of that tool and released its source code. ³Now the firm has announced that it has used its AI to predict the shapes of nearly every protein in the human body, as well as the shapes of hundreds of thousands of other proteins found in 20 of the most widely studied organisms, including yeast, fruit flies, and mice. ⁴The **breakthrough** could allow biologists from around the world to understand diseases better and develop new drugs.

⁵Proteins are made of long ribbons of amino acids, which twist themselves up into complicated knots. ⁶Knowing the shape of a protein's knot can reveal what that protein does, which is crucial for understanding how diseases work and developing new drugs---or identifying organisms that can help tackle pollution and climate change. ⁷Figuring out a protein's shape takes weeks or months in the lab. ⁸AlphaFold can predict shapes to the nearest atom in a day or two.

⁹In the new version of AlphaFold, predictions come with a confidence score that the tool uses to flag how close it thinks each predicted shape is to **the real thing**. ¹⁰Using this measure, DeepMind found that AlphaFold predicted shapes for 36% of human proteins with an accuracy that is correct down to the level of individual atoms. ¹¹"This is good enough for drug development," says DeepMind cofounder and CEO, Hassabis.

¹²Even predictions that are not fully accurate at the atomic level are still useful. ¹³For more than half of the proteins in the human body, AlphaFold has predicted a shape that should be good enough for researchers to figure out the protein's function. ¹⁴The rest of AlphaFold's current predictions are either incorrect or are for the third of proteins in the human body that don't have a structure at all until they bind with others.

5. Based on the passage, it can be inferred that a big advantage AlphaFold has over more traditional methods of predicting the shape of a protein is...
 - a. its ease of use for biologists.
 - b. the rate at which it can make predictions.
 - c. its ability to always yield fully accurate predictions.
 - d. the usefulness of its predictions.
 - e. its ability to make predictions without any kind of human intervention.
6. Which of the following statements is true according to the passage?
 - a. DeepMind announced that it has predicted every protein in humans.
 - b. For a prediction to be useful to researchers, it must be accurate down to the atomic level.
 - c. AlphaFold has always been able to estimate how accurate its predictions are.
 - d. AlphaFold can produce results that are highly accurate.
 - e. It could take months for AlphaFold to predict protein shapes in the lab.

A study of more than 57,000 childcare providers across the United States has found that those who continued working through the initial months of the pandemic weren't more likely to catch COVID-19 than those who were out of work. The findings, which were published on October 14 in the journal *Pediatrics*, indicate that childcare programs aren't likely to spread COVID-19 through the community when employees take precautions such as wearing masks and keeping the kids in their care socially distanced from each other.

"These childcare providers were doing near-Herculean things to try to keep children safe; the good news is it appears to have worked," says Walter Gilliam, director of Yale University's Edward Zigler Center in Child Development and Social Policy and a co-author of the new findings. "We

need to continue supporting childcare providers and making sure that childcare providers have what they need in order to continue doing this."

Although children do transmit the novel coronavirus to other kids and adults, the role schools and childcare centers play in spreading COVID-19 is still poorly understood. To find out whether people who work in childcare centers have an elevated risk of catching COVID-19, Gilliam and his colleagues surveyed these providers in all 50 states as well as Washington, D.C., and Puerto Rico in late May and early June. The respondents represented over 70 percent of the nations more than 3,100 counties. Roughly half had worked in childcare programs that closed near the beginning of the pandemic.

Of the 57,335 respondents, 427 (less than 1%) reported that they had tested positive for or been hospitalized with COVID-19. At the time, roughly the same amount of adults in the United States had tested positive for the novel virus. Gilliam and his colleagues found that the respondents who continued working were more likely to catch COVID-19 than those whose childcare centers had closed. However, Black, Latino, and Native American childcare workers were more likely to become infected than their white peers, regardless of whether they had been working during the first three months of the pandemic. Childcare workers in counties with high numbers of COVID-19 deaths had an elevated risk of catching COVID-19, whether or not they were working.

Respondents who'd continued working took extensive steps to minimize the risk of spreading COVID-19 in their workplaces, where most of the children they cared for were under 6 years old. More than 90 percent reported that staff and children frequently washed their hands and that indoor surfaces were disinfected at least daily; more than half disinfected indoor surfaces three times every day. Many also screened the children for a fever or other symptoms of COVID-19 and used social distancing measures such as placing seats more than 6 feet apart, keeping children in small and separate groups, and staggering arrival and departure times.

7. To put in another word, we can know that the text says that...
 - a. Childcare providers who took precautions while working during the pandemic did not have a heightened risk of catching COVID 19
 - b. Social distancing can reduce the number of COVID 19's cases
 - c. Black, Latino, and Native American childcare workers are at risk of being infected
 - d. Children transmit the novel coronavirus to other kids or adults
 - e. Only less than 1% reported being tested positive with COVID 19
8. By looking at the text, we can describe that the tone of the author is....
 - a. Sardonic
 - b. Pessimistic
 - c. Vindictive
 - d. Exuberant
 - e. Cynical

This passage is adapted from "Japanese occupation." *Encyclopaedia Britannica*.

¹During its occupation, the Japanese military authorities in Java, having interned Dutch administrative personnel, found it necessary to use Indonesians in many administrative positions, which thus gave them opportunities that had been denied them under the Dutch. ²In order to secure popular acceptance of their rule, the Japanese sought also to enlist the support of both nationalist and Islamic leaders. ³Under this policy Sukarno and Hatta both accepted positions in the military administration.

⁴Though initially welcomed as liberators, the Japanese gradually established themselves as overlords. ⁵Their policies fluctuated according to the exigencies of the Pacific War, but in general their primary object was to make the East Indies serve Japanese war needs. ⁶Nationalist leaders, however, felt able to trade support for political **concessions**. ⁷Sukarno was able to convince the

administration that Indonesian support could be mobilized only through an organization that would represent genuine Indonesian aspirations.

9. It could reasonably be inferred from the passage that ...
 - a. The Japanese gained the support of the Indonesian people gradually from the beginning of its occupation.
 - b. Before the Japanese occupation of Indonesia, there were few to no Indonesians in many administrative positions.
 - c. The policies under the Japanese rule were relatively stable.
 - d. Sukarno refused to collaborate with the Japanese.
 - e. Japan occupied Indonesia with the intention of liberating it.

These days, far too many of us end our workdays by closing one screen and turning to another. And with COVID-19 cases continuing to spike across the country, staying home and staring at various screens is admittedly one of the more responsible ways to spend your free time. But there's long been a notion that focusing your eyes---and your mind---on a device for too long can have negative effects on your health. How can we benefit from screen time without letting it destroy our sense of calm? The answer is simpler than you might think.

Weekly screen time has been going up for years. According to a 2019 report by the research nonprofit group Common Sense Media, American kids aged 8 to 12 spend about four hours and 44 minutes on screens every day. Teenagers spend an average of 7 hours and 22 minutes online---and that amount doesn't include time spent using a device for schoolwork. For adults, those numbers are even more staggering, with many surveys and studies done over the last several years showing that most of us spend the majority of our waking hours in front of screens. A 2018 Pew Research Poll found that almost one-third of American adults are online more than a handful of times a day.

In parallel with this increased time staring at a device, a number of studies have emerged to investigate the health effects of increased screen time, especially over the past decade. Some of them have found a correlation between more time spent in front of a device and worse mental health outcomes, including a 2017 study published in *Preventive Medicine Reports*, which found that adults who spent six hours or more in front of screens had a higher risk for depression. And a 2018 study, published in the same journal, found that among teenagers (aged 14 to 17 years), those who had a high use of screen time (seven or more hours a day), were more than twice as likely to have been diagnosed with depression or anxiety in the past year. The study also found that moderate screen users (those who spent about four hours a day in front of devices) also had lower psychological well-being.

But, as psychologists and other researchers studying screen time's effects on our overall mental health point out, many of these studies come with limitations. The time in front of a device or screen was self-reported, and none of these studies can draw any causative conclusions. Some experts argue that it's unclear if those with depression are simply more likely to spend more time in front of a screen. Further, a meta-analysis published in January 2020 in the *Journal of Child Psychology and Psychiatry*---which examined all studies on screen time and mental health released between 2014 and 2019---found that the conclusions were mixed, with some showing increased screen time had a negative effect on wellbeing and others showing no impact at all or even a positive effect.

A lack of quality long-term and randomized controlled trials (ones that could inch closer to establishing a causative link between screen time and mental health) makes it nearly impossible to say that increased screen time causes health problems, according to a 2020 review in *PLOS ONE*. However, the review also noted that more research should be done to better understand and identify any potential causative effects, as screens are not likely to go away anytime soon.

10. The have the lowest psychology well being because of the number of screen time
- Teenagers
 - Moderate users
 - Kids
 - Adults
 - All answers are correct
11. Which of the following is not part of the text?
- Weekly screen time has been going up these past years
 - With the occurrence of COVID-19, we have more time to look at screens
 - A 2018 Poll states that approximately one-third of American adults are online more than a handful of times a day.
 - The research and study have to face some difficulties
 - Screen time is not the root of all evil
12. The best choice to describe what this whole text talks about is...
- How daily screen affects human's well being
 - How to create screen-life balance
 - The use of screens can affect our sleep cycle
 - To what extent is the downside of screen time done
 - The increase of weekly screen time for years

For older adults, social engagement may help strengthen brain areas associated with dementia, a new study published in the *Journal of Gerontology: Psychological Sciences* finds. The investigation's conclusions have led researchers to suggest that prescribing socialization could help older adults foster brain health. "Once people lose brain cells, they're usually gone forever," says Cynthia Felix, a geriatrician at the University of Pittsburgh. That leads to dementia, a debilitating condition that affects millions of Americans each year. Treatments for some dementia symptoms exist, but there is no cure. "The better way to approach dementia, I feel, would be to prevent it," Felix says.

Felix and her colleagues relied on data from a large National Institutes of Health study that took place from the late nineties to 2011. They gathered information from 293 participants who were an average of 83 years old and still lived in community settings rather than assisted living or other facilities. The researchers were keen on using this data because the analysis included the use of a sophisticated kind of MRI that can see microstructural changes in the brain. These changes happen before cell loss occurs, says Felix. Early detection of these changes might allow doctors to prescribe socialization early on as a way to keep the regions of the brain associated with social activity active and prevent them from atrophying.

This specific type of MRI data is expensive to obtain, so it made sense to use the study data that exists rather than starting from scratch, says Felix. Participants in this study had answered a broad series of questions about their social engagement like whether they regularly visited other people and if they lived with a partner or other person, as well as a number of other metrics. Felix and her colleagues then devised a scoring system to rank the responses. Then they compared each participant's score with their MRI scans. They found that people who scored higher on the social engagement questions had, on average, more cellular integrity in the parts of the brain that relate to social interaction. They also found it didn't take much to preserve brain health: a visit with one person, once a week was enough to make a notable difference.

At this point, it's not certain whether socialization helps promote brain health or the other way around. But the relationship between these two things warrants more study, Felix says. Dementia is a significant and currently untreatable problem, and this study demonstrates that a simple intervention might be able to help people who are on the cusp of developing dementia. The

research component of this study occurred pre-pandemic, but Felix says the results are particularly important now when it's extra-hard for older adults to safely socialize. Although there isn't data about whether online activity would make as much of an impact as in-person meetups, she says, something is definitely better than nothing. It's still engaging the brain regions we need to connect with one another.

13. Below are the information depicted from the text above except...
 - a. Dementia can not be cured
 - b. Establishing socialization for older adults can improve their brain's health
 - c. The micro changes inside brain happen before the loss of the brain cells
 - d. Not all the memory loss is due to dementia
 - e. It is hard for older adults to socialize now in this severe situation

14. By reading the text above, we can say that spending time with friends and family is ____ at all stages of life
 - a. Meaningless
 - b. Salient
 - c. Insignificant
 - d. Frivolous
 - e. Trivial

15. What does the first sentence from the third paragraph do?
 - a. It states a problem
 - b. It restates an idea from the last paragraph
 - c. It gives an example
 - d. It tells an opinion
 - e. It analyzes the statement made in the first